



S. A. ENGINEERING COLLEGE, CHENNAI-77

(Autonomous-Institute Level Research Centre-Affiliated to Anna University)

Accredited by NBA, NAAC with "A" Grade and ISO 9001:2015 certified institution
Poonamallee – Avadi Main Road, Thiruverkadu, Chennai – 600 077.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

5.5. Faculty competencies in correlation to Program Specific Criteria (10)

The competency of faculty members is evaluated based on their academic qualifications, relevant training, certifications, achievements, and research publications. They effectively communicate their specialized knowledge to mentor students, helping them excel academically and prepare for various events such as Smart India Hackathons, Ideations, internships, paper presentations, and project presentations. Faculty members demonstrate continuous growth in their fields by publishing research in prestigious journals and offering consultancy services to industries. Their competence is assessed based on excellence in key areas and program-specific domains.

To measure the competency of faculty members, following factors are considered.

- Degree specialization (UG / PG /PhD),
- Research area, Research guidance (UG/PG/PhD),
- Research paper publication,
- Funded Projects Received
- Books, Book Chapters Published
- Patents Published, Products developed
- Online courses / FDP / training programs / workshops / seminars / series of webinars attended in relevance to the academic specialisation.
- Guest lecturers delivered

Knowledge Areas



Association for
Computing Machinery

Knowledge Areas of CS2023:

- Algorithmic Foundations (AL)
- Architecture and Organization (AR)
- Artificial Intelligence (AI)
- Data Management (DM)
- Foundations of Programming Languages (FPL)
- Graphics and Interactive Techniques (GIT)
- Human-Computer Interaction (HCI)
- Mathematical and Statistical Foundations (MSF)
- Networking and Communication (NC)
- Operating Systems (OS)
- Parallel and Distributed Computing (PDC)
- Security (SEC)
- Society, Ethics, and the Profession (SEP)
- Software Development Fundamentals (SDF)
- Software Engineering (SE)
- Specialized Platform Development (SPD)
- Systems Fundamentals (SF)

IEEE-CS



IEEE
COMPUTER
SOCIETY

AAAI



 Association for Computing Machinery ACM CS2023 Knowledge Area (KA)	 CSE Department Domain
AI – Artificial Intelligence	Artificial Intelligence, Machine Learning, Data Science
AL – Algorithmic Foundations	Data Structures & Algorithm Analysis, Programming for Problem Solving
DM – Data Management	Data Science, Cloud Computing
MSF – Mathematical & Statistical Foundations	Machine Learning, Data Science
NC – Networking & Communication	Wireless Networks, Internet of Things
SEC – Security	Cyber Security ,Cloud Computing
OS – Operating Systems	Operating System
SF – Systems Fundamentals	Networking ,Compiler Design, Cyber Security
SPD – Specialized Platform Development	Internet of Things, Cloud Computing
SEP – Society, Ethics & Profession	Cyber Security, Artificial Intelligence
AR – Architecture & Organization	Internet of Things, Operating System
HCI – Human-Computer Interaction	Programming for Problem Solving, Internet of Things

Table: Summary of Faculty competencies in correlation to Program Specific Criteria

FACULTY NAME	Machine learning	Wireless Networks	Artificial intelligence	Cloud Computing	Internet of Things	Compiler Design	Cyber Security	Program for Problem Solving	Data Structures & Algorithm Analysis	Data Science
Dr.R.GEETHA	Y	Y	Y							
Dr.M.ANURADHA	Y	Y	Y							
Dr.D.HEMANAND		Y	Y	Y	Y					
Dr.S.SIVAKUMAR		Y			Y	Y				
Dr. ANTONIDOSS A	Y		Y	Y						
Dr.D.CHITRA DEVI	Y	Y		Y						
Dr.N.THILAGAVATHI	Y				Y					
Dr.V.KALIRAJ	Y		Y							
Mr.C.BALAKRISHNAN	Y						Y			
Mr.A.MANI				Y	Y			Y		
Mr.M.BALASUBRAMANIAN	Y						Y	Y		
Mr.S.MUTHUKUMARASAMY		Y			Y					
Mrs.V. SARASWATHI	Y				Y			Y		
Mrs.L.SUDHA				Y					Y	
Mrs.S.ARUMAI SHINEY	Y		Y		Y					
Mrs.V.SUREKA				Y					Y	
Mrs. K.B.ARUNA				Y					Y	
Mrs.J.SANGEETHA				Y	Y	Y				
Mrs.S.GUNANANDHINI	Y							Y		
Mrs.C.LAKSHMIPRIYA	Y				Y			Y		
Mrs.V.KAVITHA							Y			
Mrs.R.SHOBANA	Y		Y				Y	Y		Y
Mrs.BELSHIA JEBAMALAR	Y						Y			
Mr.R.NITHYANANDHAN	Y			Y						
Mrs.S.GAYATHRI	Y		Y		Y			Y		
Mrs. S. N. ANANTHI	Y		Y				Y			

Mrs.N.R.G.SREEVANI								Y	Y	
Mrs.RUBASUDHA									Y	
Mrs. M.KAVITHA										Y
Ms.M.DHIVYABHARATHI				Y				Y		
Ms.G.SARANYA									Y	
Mrs.S.T.PREETHI							Y			
Mrs.O.VASANTHA KUMARI						Y				
Mrs.K. SIVASAKTHI			Y		Y					
Mrs.L.DHARANI					Y					
Mrs.M.REVATHI			Y							
Mrs.A.SHAINI								Y		
Mrs.M.SHYNI		Y			Y					Y

MACHINE LEARNING



Dr.R.GEETHA
Professor & HEAD



Dr.M.ANURADHA
Professor



Dr ANTONIDOSS A
PROFESSOR



Dr.N.THILAGAVATHY
Associate professor



Dr.V.KALIRAJ
Associate professor



Mr.C.BALAKRISHNAN
Associate professor



Mr.M.BALASUBRAMANIAN
Associate professor



Mrs.V.SARASWATHI



Mrs.S.ARUMAI SHINEY



Mrs.S.ANANTHI



Mrs.S.GUNANANDHINI



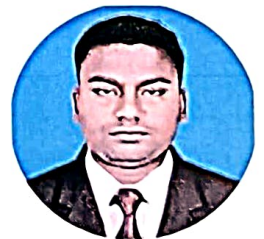
Ms.C.LAKSHMI PRIYA



Ms.R.SHOBANA



Ms.BELSHIA JEBAMALAR



Mr.R.NITHYANANDHAN



Ms.S.GAYATHRI



Dr. D CHITHRA DEVI

WIRELESS NETWORKS



DR.R.GEETHA
Professor & HEAD



DR.M.ANURAHA
Professor



DR.S.SIVAKUMAR
professor



DR.D.HEMANNAND
PROFESSOR



DR.D.CHITRA DEVI
associate professor



MR.S.MUTHUKUMARASAMY
Associate professor

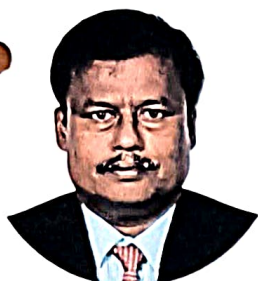
ARTIFICIAL INTELLIGENCE



Dr.R.GEETHA
PROFESSOR & HEAD



Dr.D.HEMANAND
PROFESSOR



Dr ANTONIDOSS A
PROFESSOR



Dr.M.ANURADHA
PROFESSOR



Dr.V.KALIRAJ
ASSOCIATE PROFESSOR



Ms.S.GAYATHRI
ASSISTANT PROFESSOR



Mrs.S.ARUMAI SHINEY
ASSISTANT PROFESSOR



Ms.SHOBANA
ASSISTANT PROFESSOR



Mrs.K.SIVASAKTHI
ASSISTANT PROFESSOR



Mrs.S.N.ANANDHI
ASSISTANT PROFESSOR



Mrs.M.REVATHY
ASSISTANT PROFESSOR

CLOUD COMPUTING



Dr. D HEMANAND
PROFESSOR



Dr. D CHITHRA DEVI
ASSOCIATE PROFESSOR



Dr ANTONIDOSS A
PROFESSOR



Mr.A.MANI
ASSOCIATE PROFESSOR



Mrs.L.SUDHA
ASSISTANT PROFESSOR



Dr.V.SUREKA
ASSISTANT PROFESSOR



Mrs.K.B..ARUNA
ASSISTANT PROFESSOR



Mrs.J.SANGEETHA
ASSISTANT PROFESSOR



Mr.R.NITHIYANANDHAM
ASSITANT PROFESSOR



Mrs.R.DIVYA BHARATHI
ASSISTANT PROFESSOR

INTERNET OF THINGS



Dr.D.HEMANAND
professor



Dr.S.SIVAKUMAR
professor



Dr.N.THILAGAVATHY
Associate professor



Mr.A.MANI
Associate professor



Mr.S.MUTHUKUMARASAMY
Associate professor



Mrs.V.SARASWATHI



Ms.S.GAYATHRI



Mrs.S.ARUMAI SHINEY



Mrs.J.SANGEETHA



Ms.LAKSHMI PRIYA



Mrs.K.SIVASAKTHI



Mrs.L.DHARANI



COMPILER DESIGN



Dr.S.SIVAKUMAR
Professor



Mrs.J.SANGEETHA
Assistant professor

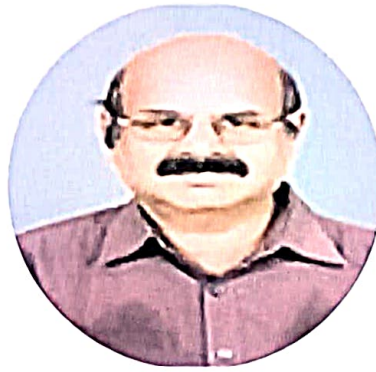


Mrs.VASANTHAKUMARI
Assistant Professor

CYBER SECURITY



Mr.M.BALASUBRAMANIAN
Associate professor



Mr.C.BALAKRISHNAN
Associate professor



Ms.V.KAVITHA



Ms.SHOBANA



Ms.BELSHIA JEBAMALAR



Mrs.S.N.ANANTHI



Mrs.S.T.PREETHI

PROGRAM FOR PROBLEM SOLVING



Mr.M.BALASUBRAMANIAN
Associate professor



MRS.V.SARASWATHI
Assistant Professor



MR.A.Mani
Associate professor



MRS.A.SHAINI
Assistant Professor



MS.S.GAYATHIRI
Assistant Professor



MRS.S.GUNANANDHINI
Assistant Professor



MRS.LAKSHMI PRIYA
Assistant Professor



MRS.N.R.G.SREEVANI
Assistant Professor



MRS.R.DHIYA BHARATHI
Assistant Professor



MS.SHOBANA
Assistant Professor

DATA STRUCTURE & ALGORITHM ANALYSIS



Mrs.L.SUDHA
Assisant professor



Mrs.v.SUREKA
Assisant professor



Mrs.k.B.ARUNA
Assisant professor



Mrs.N.R.G.SREEVNI
Assisant professor



Mrs.RUBASUDHA
Assisant professor



Mrs.G.SARANYA
Assisant professor

DATA SCIENCE



Mrs.M.KAVITHA
Associate Professor



Ms.SHOBANA
Assistant Professor

Program Specific Criteria : MACHINE LEARNING

Name of the Faculty	Competency Attained Through
Dr.R.GEETHA	- Published a paper on topic " Human Stress Detection in and Through Sleep Patterns Using Machine Learning Algorithms" - Done the Consultancy Project on topic "National Bonds Thrift Application" - Published patent on topic "Generative AI-Driven Credit Card Fraud Detection System with Block Chain based Cybersecurity Enhancement" - Resource person for FDP titled "Machine Learning in Science and Engineering Models and Applications" S.A. Engineering College with E & ICT Academy, NIT Warangal - Delivered Guest Lectures /Invited Talks on "Machine Learning algorithms, IIT Delhi" - Handled the subject "Machine learning" more than 5 times. - Completed NPTEL Course Related to Machine Learning - Guiding the Research Scholar in Machine Learning Domain
Dr.M.ANURADHA	- Published a paper on topic ""Resource allocation scheme for CCRN using hybrid Giza Pyramids construction-based complex-valued satin bowerbird optimization" - Published a patent on the title "Design and Implementation of IOT Based Weather Monitoring System"
Dr. A.ANTONIDOSS	- Published a paper on topic " Optimizing Creditworthiness Assessment: Leveraging PCA and ESVM for Improved Credit Score Analysis" - Published a paper on topic " Deep Learning-Based Monitoring Sustainable Decision Support System for Energy Building to Smart Cities with Remote Sensing Techniques" - Handled the subject "Machine learning" more than 5 times.

	Received the Award: • TEC- Best Faculty Award for the year 2016-17 ,(TEC - Tagore Engineering College) • Global Faculty Award 2020 (AKS AWARDS) https://gfa.aksawards.com/AwardWinners?date=2020 • Global Faculty Award 2021 (AKS AWARDS) https://gfa.aksawards.com/AwardWinners?date=2021 • Encouragement Award - Publications - 2022, Hindustan University
Dr.D.CHITRA DEVI	• Published a paper on the title "Novel Beam Sector Antenna-Based Routing Protocol for MANET Hidden and Exposed Terminal Problem" • Published a paper on the title "A comprehensive evaluation of machine learning algorithms for precise energy consumption forecasting in smart homes"
Dr.N.THILAGAVATHI	• Attended FDP on topic "Introduction to Machine Learning" • Completed Certification on topic "MATLAB, UHV1,UHV2"
Dr.V.KALIRAJ	• Published Paper on the title "A design of movie script generation based on natural language processing by optimized ensemble deep learning with heuristic algorithm"
Mr..M.BALASUBRAMANIAN	• Published a paper on the title "COVID Detection: deep convolutional neural networks-based automatic detection of COVID-19 with chest x-ray images" • Published a patent on topic "Multiple Disease Prediction using Machine Learning Algorithms" • Completed MATLAB Certificate on Machine Learning

Mrs.V. SARASWATHI	- Published paper on the title "Vehicular Assistance Using ML Algorithm, Implementation of Hyper Parameter Optimization in Liver Disease Prediction" - Attended FDP on topic "Machine Learning Algorithms and its Applications"
Mrs.S.ARUMAI SHINEY	- Published paper on topic " AGBUNet: an enhanced CNN-UNET architecture for the prediction of above ground biomass using deep learning - Leveraging Deep Learning Models for Targeted Above ground Biomass Estimation in Specific Regions of Interest"
Mrs.C.LAKSHMIPRIYA	- Published paper on topic " Meta-heuristics and machine learning applications in complex systems" - Completed NPTEL on topic "NATE" - Done "Wipro Faculty Trainer (Java Full Stack)" - Published paper on topic " Enhancing Intrusion Detection Systems with Machine Learning" - Published patent on the title "Automated Student Attendance System Using Real-Time Face Detection and Recognition" - Handled "Java programming " more than 5 times.
Mrs.S.GAYATHRI	- Published paper on topic "A Comprehensive Survey of Machine Learning: Advancements, Applications, and Challenges" - Published patent on topic "Automated Student attendance system using Real time Face detection and recognition" - Published paper on topic "Predictive Analytics for Heart Disease Detection: A Machine Learning Approach"
Mrs.R.SHOBANA	- Attended FDP on topic "Machine Learning in Science and Engineering: Models" - Attended FDP on topic "Cloud Computing with Machine Learning"
Mrs. S. N. ANANTHI	Published paper on topic " Effective Street View Traffic Sign detection and Recognition Using Deep learning

Mr.C.BALAKRISHNAN	- Published the paper on topic "Optimizing Machine Learning Algorithms for Heart Disease Prediction in Healthcare: a Comparative Study" - Published the patent on topic "Machine Learning Based System For Correcting Improperly White-Balanced Images"
Mr.R.NITHYANANDAN	- Published paper on topic "Machine Learning Approaches for Liver Disease Prediction: A Comparative Analysis - Published paper on topic "Enhancing Healthcare Data Security in the Cloud: Integrating ML-Based Intrusion Detection Systems
Mrs.BELSHIA JEBAMALAR	- Published paper on topic " Prediction and Comparison of ML Algorithm for Heart Disease " - Attended FDP Related to Machine Learning Algorithm
Mrs.M.KAVITHA	- Published paper on topic " Classical clustering technique for segmentation of skin cancer image" - Published a book Chapter on topic "Fuzzy Clustering Technique for Segmentation on Skin Cancer Dermoscopic Images" - Published paper on topic "Enhanced Clustering Technique for Segmentation on Dermoscopic Images" - Completed NPTEL course "Introduction to Machine Learning"
Mrs.GUNANANDHINI	- Published paper on topic "Human Stress Detection in and Through Sleep Patterns Using Machine Learning Algorithms" - Published paper on topic "Integrated SPAM detection for multilingual emails" - Attended FDP related to Machine Learning

Program Specific Criteria :WIRELESS SENSOR NETWORK	
Name of the Faculty	Competency Attained Through
Dr.R.GEETHA	- Published a paper on topic" A Scalable chain framework for user identity management in a Decentralized network" - Published a paper on topic" Dynamic analysis of the e-SITR model for Remote wireless sensor network with noise and Sokol-howell functional response - Published a paper on topic" Analysis of delay aware worm propagation model in wireless IoT Systems with Ratio -dependent functional response - Attended FDP related to Wireless Sensor Networks. - Handled the subject like Cryptography and Network Security, Artificial Intelligence, Computer Networks more than 5 times - Guiding the Research Scholar in Wireless Sensor Network Domain
Dr.M.ANURADHA	- Published a paper "HRF-ExGB: Hybrid random forest-extreme gradient boosting for mobile edge computing" - Published patent on topic "An RSSI-Powered Approach to Maritime Border Surveillance" - Delivered Guest Lectures /Invited Talks. on topic "Blooms taxonomy" - NPTEL Certification on topic " Wireless Networks" - Done the certification "Introduction to Cyber Security" - Handled the subject "Computer Networks" more than 5 times " - Completed NPTEL on topic "Outcome Based Education" - Completed Certification " UHV1, UHV2" - Published book chapter on topic "Data Structures,2022 IO press"
Dr.D.HEMANAND	- Published a patent " Home Automation System with Voice Recognition for The Person with Disabilities" - Completed NPTEL on topic "Cyber Security and Privacy" - Book chapter published "Futuristic Trends in Network & Communication Technologies/2024/IIP" - Delivered Guest Lectures /Invited Talks on" AR/VR Research at SSEC, Coimbatore" - Handled the subject like Computer Networks, Cryptography and Network Security more than 5 times

	Guiding the Research Scholar in Wireless Sensor Network Domain
Dr.D.CHITRA DEVI	Published a paper "Enhancing MANET Security: A Watch Dog Routing Algorithm Approach for Intruder and Black Hole Attack Detection"
Mr.S.MUTHUKUMARASAMY	Published a paper on topic "Energy-efficient coverage in wireless sensor networks based on stacked contractive auto encoder with Manta Ray foraging optimisation algorithm"
Mrs.NRG SREE VANI	Published paper on topic "Integrated photonic devices for cancer detection"
Mrs.M.SHYNI	- Published paper on topic "Detection Monitoring Of Secure Packet Transfer Over Network Traffic" - Published paper on topic "Towards Securing Data Management in Social Networks Using Trust Based Mechanism" - Published paper on topic "Advanced Assessing Risk Analysis of Integrated Data Attacks against Power System State Estimation"

Program Specific Criteria : ARTIFICIAL INTELLIGENCE

Name of the Faculty	Competency Attained Through
Dr.R.GEETHA	• Done the Consultancy Project on topic "Cyber Security" • Published the patent on the title "An Early Cotton Plant Disease Detection Model Using Explainable AI Technique" • Resource person for "Acted as a resource person in FDP at Vel Tech High Tech Engineering College titled A "Drafting and publishing Research articles" • Delivered Guest Lectures /Invited Talks. on topic "Deployment of Artificial Intelligence and Data Science in Developing Electrical Engineering Applications-Baharat Institute of Higher Education and Research" • Handled the subject "Artificial Intelligence" more than 5 times • Guiding the Research Scholar in Artificial Intelligence Domain
Dr.M.ANURADHA	• Attended FDP on topic "Exploring the Frontiers of Artificial Intelligence: Theory, Application & Ethics"
Dr.D.HEMANAND	• Done the Consultancy Project on topic "Learning Management Software" • Published the patent on the title "Automatic LPG Gas Leakage Detection and Protection with an Alert System" • Attended FDP on topic "AI Insights: Predictive Power of MI, DL&NLP" • Guiding the Research Scholar in Artificial Intelligence Domain
Dr. A.ANTONIDOSS	• Published a book titled" Doorstep Python "https://www.amazon.in/Doorstep-Python-C-Karthikeyan/dp/1639577491). • Published a paper on topic "Artificial intelligence techniques for phishing detection" • Published a paper on topic "Adaptive segmentation from computerized tomography images using threshold". • Guided the Research Scholar in Artificial Intelligence Domain
Dr.V.KALIRAJ	• Published a paper titled "An implementation of personalized therapy in Clinical Decision Support System using adaptive transformer and hybrid deep learning network." • Published patent on the title " Smart plant monitoring System" • Delivered Guest Lectures on topic " IEEE Session Chair"
Mrs.R.SHOBANA	• Attended FDP on topic "Artificial Intelligence" • Attended FDP on topic "Exploring the frontier of Artificial Intelligence: Theory, Application and Ethics" • Handled "Artificial intelligence" twice and "Machine learning laboratory" twice

Mrs.S.ARUMAI SHINEY	• Attended FDP on topic " Exploring The Frontiers of Artificial Intelligence: Theory, Application, and Ethics"
Mrs.K. SIVASAKTHI	• Published Patent on topic "AI-Driven Personalized Cognitive Enhancement System" • Attended FDP on topic "Exploring Emerging Frontiers in Generative AI and Quantum Computing" • Handled the subject "Programming In C" more than 5 times
Mrs.M.REVATHI	• Published patent on topic "3D Image Modelling For Fashion Designing" • Attended FDP on topic "Full Stack Data Science Generative AI"
Mrs.S.GAYATHRI	• Attended FDP on topic "Exploring The Frontiers Of Artificial Intelligence: Theory, Application, And Ethics" • Completed certification on Infosys Springboard-AI, Java
Mrs. S. N. ANANTHI	• Attended FDP on topic " "Exploring the Frontiers of Artificial Intelligence: Theory, Application & Ethics"
Mrs. M.KAVITHA	• Published paper on topic "Artificial Intelligence Clustering Techniques on Dermoscopic Skin Lesion Images"

Program Specific Criteria : CLOUD COMPUTING

Name of the Faculty	Competency Attained Through
Dr.D.HEMANAND	• Delivered Guest Lectures /Invited Talks. "Text Classification and Clustering. • Handled the subject "DBMS" more than 5 times • Guiding the Research Scholar In Cloud Computing Domain
Dr. A.ANTONIDOSS	• Published a paper on topic "A new methodology for secured inventory management by average fitness-based colliding bodies optimization integrated with block chain under cloud" • Published a paper on topic "Algorithmic Analysis on Spider Monkey Local Leader-based Sea Lion Optimization for Inventory Management Integrated with Block Chain in Cloud" • Published a paper on topic "Real-time hierarchical sensitivity measure-based access restriction for efficient data retrieval in cloud" • Guided the Research Scholar in Cloud Computing Domain
Dr.D.CHITRA DEVI	• Involved in Consultancy project "TG Cyber Labs Learner's Portal" • Published a patent titled in "Specks: A Navigation Aid for the Visually Impaired Using Ultrasonic Sensors"
Mr.A.MANI	• Published paper on topic" A proficient resource allocation using hybrid optimization algorithm for massive internet of health things devices contemplating privacy fortification in cloud edge computing environment " • Published paper on topic "Cryptography in the cloud: securing cloud data with encryption"
Mrs.L.SUDHA	• Published paper on topic " Dynamic task offloading and collaborative task execution using three tier edge cloud computing (T2EC2) system for autonomous vehicles" • Published patent on the title " A Lightweight Attribute Role-Based Encryption in Mobile Cloud Environment" • Attended 21 days international FDP on Data Science Tools, National level FDP on Recent trends in cloud computing. • Completed NPTEL Certification on "Google Cloud Computing Foundations". • Attended National Conference on An Efficient DDOS TCP Flood Attack Detection and Prevention System in a Cloud Environment course era - Google Cloud Platform Fundamentals: Core

Mrs.V.SUREKA	- Published paper on topic " Dynamic task offloading and collaborative task execution using three tier edge cloud computing (T2EC2) system for autonomous vehicles" - Published patent on the title " A Lightweight Attribute Role-Based Encryption in Mobile Cloud Environment" - Attend 21 days international FDP on Data Science Tools, National level FDP on Recent trends in cloud computing - Done the NPTEL Certificate on the topic " Google Cloud Computing Foundations" - Attend National Conference on An Efficient DDOS TCP Flood Attack Detection and Prevention System in a Cloud Environment course era - Google Cloud Platform Fundamentals: Core Infrastructure
Mrs. K.B.ARUNA	- Published paper on topic " Development of hybrid heuristic-assisted dynamic data auditing strategy in cloud sector". - Published paper on the topic "Cutting-edge Facial Authentication System in MATLAB (CFAS-M)" - Completed NPTEL certificate course on the topic " Google Cloud Computing Foundations, cloud computing"
Mrs.J.SANGEETHA	Published paper on topic " Revolutionizing Image Processing: Innovations Impacts and Future Trends"
Ms.M.DIVYABHARATHI	Attended FP on Cloud Infrastructures(AWS)
Mr.R.NITHYANANDAN	Attended FDP on topic "Computer Vision and Image Processing with Intelligent Cloud Resource"

Program Specific Criteria : INTERNET OF THINGS	
Name of the Faculty	Competency Attained Through

Dr.N.THILAGAVATHI	• Attended FDP on topic "FDP on IoT - Resource person"
Mr.A.MANI	• Published paper on topic "A Smart and Reliable Health Monitoring System for Covid-19 using Internet of Things" • Attended FDP on topic "Introduction to Industry 4.0 and Industrial Internet of Things," Foundation of Cloud IoT Edge ML" • Completed Coursera Certificate related to Internet of Things.
Dr.D.HEMANAND	• Attended FDP on topic "Deep Learning and IoT driven Smart Applications" • Guiding the Research Scholar in "Internet of Things" Domain
Mrs.S.ARUMAI SHINEY	• Published patent on the title " Automated Water Tank Cleaning And Monitoring Device With IOT"
Mrs.V. SARASWATHI	• Published paper on topic "IoT Based Vehicle Tracking and Monitoring System Using Smart Application" • Published Patent on the title "Remote Controlled Gas Knob Rotating Systems" • Book Published on the title "Microprocessor and Micro Controller ,June 2019,Charulatha publications"
Mrs.K. SIVASAKTHI	• Handled the subject "Design and Analysis of Algorithms:" more than 5 times
Mr.S.MUTHUKUMARASAMY	• Published Patent on topic "Smart Healthcare Systems Via IOT"
Mrs.L.DHARANI	• Published the paper on topic "Internet of Things Based on Electrical-Device Reconnaissance And Controller method". • Attended FDP on the topic "IoT" • Completed NPTEL certificate on the topic "IoT"
Mrs.S.GAYATHRI	• Published patent on topic "Smart Agriculture Monitoring System based on IoT"
Mrs.C.LAKSHMIPRIYA	• Published patent on topic "Smart Agriculture Monitoring System based on IoT"
Mrs.J.SANGEETHA	• Published patent on the title " Smart Agriculture Monitoring System based on IOT"
Mrs.M.SHYNI	• Published paper on topic "Smart Farming Solutions for cost effective agriculture using IoT"

Program Specific Criteria: COMPILER DESIGN	
Name of the Faculty	Competency Attained Through
Dr.S.SIVAKUMAR	• Attended FDP related to Compiler Design • Handled the subject "Compiler design" more than 5 times
Mrs.J.SANGEETHA	• Attended FDP related to Compiler Design • Handled the subject "Compiler design " more than 5 times
Mrs.O.VASANTHA KUMARI	• Handled the subject "Compiler design" more than 5 times • Attended FDP on topic "Transforming Education: Integrating OBE and NEP 2020(Two Weeks)"

Program Specific Criteria: CYBER SECURITY	
Name of the Faculty	Competency Attained Through

Dr.R.GEETHA	Guiding Ph.D. scholar on "Cyber Security"
Dr.D.HIEMANAND	Done the certification Introduction to Cyber Security
Mr.C.BALAKRISHNAN	- Completed NPTEL certification on topic "Cyber Security and Privacy" - Handled the subject "Cyber Security" more than 5 times. - Attended FDP on topic "One week FDP on Nano Computing" - Completed "MATLAB" certification
Mr.M.BALASUBRAMANIAN	- Published Patent on the title "Online Voting System Using Block chain" - Done the Consultancy Project "Volumetric Segmentation using Monai"
Mrs. S. N. ANANTHI	- Published paper on topic "Enhanced Cybersecurity: Development and Evaluation of a secure Data Sharing Paradigm Based on Block chain Technology" - Published patent on topic "Smart Plant Monitoring system" - Completed NPTEL certification on topic "Cyber Security and Privacy" - Complete the NPTEL certification on topic "Outcome Based Education" - Published patent on the title "Smart Agriculture using Block Chain Application for Tagging Farm Producers"
Mrs.BELSHIA JEBAMALAR	- Completed NPTEL certification on "Cyber Security and Privacy" - Attended FDP related to Cyber Security
Mrs.S.T.PREETHI	Attended FDP on topic "Enhancing Cyber Threat Detection Through Incorporating Digital Technologies in ATAL"
Mrs.R.SHOBANA	- Published paper on topic "Cybersecurity: Strategies, Sector-Specific Challenges, and Future Prospects" - Completed NPTEL certification on "Cyber Security and Privacy" - Done "Introduction to Cybersecurity Foundations" by course era
Mrs.KAVITHA V	- Published paper on topic "Cyber-Attack Prediction in Virtual IoT using Leveraging Mechanism" - Attended FDP related to Cyber Security

Program Specific Criteria: PROGRAMMING FOR PROBLEM SOLVING

Name of the Faculty	Competency Attained Through
---------------------	-----------------------------

Mr.A.MANI	- Published patent on topic "Face Detection and Recognition for Criminal Identification System" - Attended FDP related to Programming Papers - Handled Java, Python++ and C with lab are more than 3 Times
Mr.M.BALASUBRAMANIAN	- Completed MATLAB Certificate on OOP - Attended FDP Related to Programming Papers - Handled python Programming subject more than 5 times. - Handled Data base Management subject more than 5 times.
Mrs.V. SARASWATHI	- Published patent on titled " Piggy Bank Using Micro python Along with Firebase" - FDP attended on topic " Data Science and its Tool" - Completed NPTEL Certificate on topic " Python for Data Science" - Completed "Udemy Certificate course for "Python Programming" - Guest Speaker for the seminar titled "R Programming" - Handling subject like Python Programming, C Programming, Internet Programming for more than 5 times - Participated in FDP on "Amazon Web Services (AWS)" - Done the Coursera certificate for "Validate HTML Forms with JavaScript and HTML"
Mrs.C.LAKSHMIPRIYA	- Published paper on topic " Translation System for the Visually Impaired from English to Braille" - Handled Programming in C subject more than 5 times.
Mrs.R.SHOBANA	- Done the Infosys springboard on "Java programming" and "OOPS using C++" by ICT Academy - Handled Java, C++ and C with labs.
Mrs.S.GAYATHRI	- Published paper on topic " A Survey on Natural Language Processing and its Applications" - Attended FDP on the topic "The Art of Research and Development"
Mrs.NRG.SREEVANI	Completed UHV1 Course
Ms.M.Dhivyabharathi	Attended FDP on topic " Python Programming"
Mrs.S.ARUMAI SHINEY	- Published Paper on topic "verification of decentralized delegation using aggregate signature" on the domain UI/UX. - Done the NPTEL certification on the topic " Machine Learning"
Mrs.S.GUNANANDHINI	- Completed UHV Course - Attended FDP related to Programming subjects - Programming papers are handled more than 2 times

Ms.G.SARANYA	• Completed UHV Course • Attended FDP related to Programming subjects • Programming papers are handled more than 2 times
Mrs.A.SHAINI	• Completed UHV Course • Attended FDP related to Programming subjects

Program Specific Criteria: DATA STRUCTURES AND ALGORITHM ANALYSIS	
Name of the Faculty	Competency Attained Through
Mrs.L.SUDHA	• Handling "Data Structures" subject more than 5 times • Published paper on topic " Nature Inspired Meta-Heuristic Optimization Algorithms Capitalized"

	Completed the Course era - Overview of Data Visualization Online Webinar on "Importance of Fibonacci Series"
Mrs.V.SUREKA	- Published paper on topic "Nature Inspired Meta-Heuristic Optimization Algorithms Capitalized" - Completed Course era - Overview of Data Visualization Online Webinar on "Importance of Fibonacci Series"
Mrs. K.B.ARUNA	- Published paper on topic "Nature Inspired Meta-Heuristic Optimization Algorithms Capitalized" - Published paper on the topic "Vehicle Insurance Claim Prediction"
Mrs.RUBASUDHA	- Attended FDP on topic "Programming and Data Structures II" - Completed MATLAB Courses
Mrs.NRG.SREEVANI	Handling "Data Structures" subject more than 5 times

Program Specific Criteria: DATA SCIENCE	
Name of the Faculty	Competency Attained Through

Mrs.R.SHOBANA	- Published Patent on topic " An Enhanced Prediction Technique Of Air Pollution Using Data Science" - Attended FDP on topic "Fundamentals of Data Science and Analytics", - Attended FDP on topic "Data Science tools" - Attended FDP on topic "Research Methodology & Data Analysis"
Mrs. K.B.ARUNA	Published patent on the topic " An Enhanced Prediction Technique Of Air Pollution Using Data Science"
Mrs. M.SHYNI	Attended FDP on the topic "Data Science with Python"
Mrs.K.SIVASAKTHI	- Published paper on topic "Financial Data Security Using Block Chain and Proxy Re Encryption" - Published patent on the title "3d Image Modelling for Fashion Designing" - Attended FDP on topic " Full Stack Data Science Generative AI"

FACULTY NAME: Dr.R.Geetha

Domain Name: Machine Learning, Wireless Network, Artificial intelligence

Program Specific Criteria: MACHINE LEARNING

Competency Attained Through

- Published a paper on topic " Human Stress Detection in and Through Sleep Patterns Using Machine Learning Algorithms"
- Done the Consultancy Project on topic "National Bonds Thrift Application"
- Published patent on topic "Generative AI-Driven Credit Card Fraud Detection System with Block Chain based Cybersecurity Enhancement"
- Resource person for FDP titled "Machine Learning in Science and Engineering Models and Applications" S.A. Engineering College with E & ICT Academy, NIT Warangal
- Delivered Guest Lectures /Invited Talks on "Machine Learning algorithms, IIT Delhi"
- Handled the subject "Machine learning" more than 5 times.
- Completed NPTEL Course Related to Machine Learning
- Guiding the Research Scholar in Machine Learning Domain

Program Specific Criteria : WIRELESS SENSOR NETWORK

Competency Attained Through

- Published a paper on topic" A Scalable chain framework for user identity management in a Decentralized network"
- Published a paper on topic" Dynamic analysis of the e-SITR model for Remote wireless sensor network with noise and Sokol-howell functional response
- Published a paper on topic" Analysis of delay aware worm propagation model in wireless IoT Systems with Ratio -dependent functional response
- Attended FDP related to Wireless Sensor Networks.
- Handled the subject like Cryptography and Network Security, Artificial Intelligence, Computer Networks more than 5 times
- Guiding the Research Scholar in Wireless Sensor Network Domain

Program Specific Criteria :ARTIFICIAL INTELLIGENCE

Competency Attained Through

- Done the Consultancy Project on topic "Cyber Security"
- Published the patent on the title "An Early Cotton Plant Disease Detection Model Using Explainable AI Technique"
- Resource person for "Acted as a resource person in FDP at Vel Tech High Tech Engineering College titled A "Drafting and publishing Research articles"
- Delivered Guest Lectures /Invited Talks. on topic "Deployment of Artificial Intelligence and Data Science in Developing Electrical Engineering Applications-Baharat Institute of Higher Education and Research"
- Handled the subject "Artificial Intelligence" more than 5 times
- Guiding the Research Scholar in Artificial Intelligence Domain

FACULTY NAME: Dr. MANU RADHA

Domain Name: Machine learning, Wireless Network, Artificial Intelligence

Program Specific Criteria: MACHINE LEARNING

Competency Attained Through

- Published a paper on topic "Resource allocation scheme for CCRN using hybrid Giza Pyramids construction-based complex-valued satin bowerbird optimization"
- Published a patent on the title "Design and Implementation of IOT Based Weather Monitoring System"

Program Specific Criteria :WIRELESS SENSOR NETWORK

Competency Attained Through

- Published a paper "HRF-ExGB: Hybrid random forest-extreme gradient boosting for mobile edge computing"
- Published patent on topic "An RSSI-Powered Approach to Maritime Border Surveillance"
- Delivered Guest Lectures /Invited Talks. on topic "Blooms taxonomy"
- NPTEL Certification on topic " Wireless Networks"
- Done the certification "Introduction to Cyber Security"
- Handled the subject "Computer Networks" more than 5 times "
- Completed NPTEL on topic "Outcome Based Education"
- Completed Certification " UHV1, UHV2"
- Published book chapter on topic "Data Structures,2022 IO press"

Program Specific Criteria :ARTIFICIAL INTELLIGENCE

Competency Attained Through

- Attended FDP on topic "Exploring the Frontiers of Artificial Intelligence: Theory, Application & Ethics"
- Guiding the Research Scholar in Artificial Intelligence Domain